TrialGuard AI
Clinical Trial Risk Monitor & Protocol Deviation Detector

Team Name: TrialGuard Squad

Team Members: Daksh Soni (Lead), Visham, Het, Manav

Event: IBM Bob AI Innovation Hackathon 2026 (AI Track)

Application: Standalone Full-Screen Web Application (React + FastAPI)

Tagline: "Detect Earlier. Prioritize Smarter. Act Faster."

Positioning: Enterprise Clinical Decision-Support Prototype using Synthetic Trial Telemetry

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The Clinical Trial Monitoring Challenge
Manual site monitoring creates operational blind spots and delayed risk discovery

Patient Records + Visit Telemetry + Site Data
↓ Data ingestion
Large Volume of Multi-Site Clinical Data
↓ Manual review bottleneck
Hard-to-Detect Protocol Deviations: Late Visits, Dosing Errors, Prohibited Meds
↓
Delayed Risk Identification & Reactive Site Audits

High telemetry volume creates manual inspection fatigue; delayed audit signals hide deviations; sequential review lacks explainable site-level risk scoring.

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Introducing TrialGuard AI
AI-assisted clinical trial monitoring for real-time risk intelligence

TrialGuard AI evaluates clinical trial records against protocol rules, detects deviations, classifies severity, calculates an explainable 0-100 site risk score, and assists in generating standardized CAPA reports.

Workflow: Clinical Protocol → Patient Data → Protocol Comparison → Deviation Detection → Severity Rating → Site Risk Score → AI Copilot Explanation → CAPA Report

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What TrialGuard AI Does
Six core capabilities for proactive clinical trial risk oversight

1. Clinical Trial Dashboard — Patients, sites, active deviations, top critical risk sites.

2. Protocol Deviation Detector — Missed visits, late visits, incorrect dosing, prohibited meds, missing labs.

3. Severity Classification — Major, Minor, Administrative.

4. Explainable Site Risk Score — Weighted 0-100 scoring with transparent factor attribution.

5. IBM Bob AI Copilot — Database-grounded conversational assistant for telemetry, risk and CAPA queries.

6. CAPA Report Generator — Issue summaries, root-cause analysis, corrective actions and PDF export.

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Technical Architecture
Full-stack decoupled Web Application with grounded AI telemetry

Presentation: React 18 + TypeScript + Tailwind CSS + Recharts
↓ REST API (JSON)
Backend: FastAPI + Python + Pydantic
↓
SQLite + SQLAlchemy ORM | Rule Engine / Deviation Detector | AI Copilot Layer
AI Copilot: watsonx.ai REST + Deterministic Fallback
↓
CAPA Report Generator → PDF Export / DB Record

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From Trial Data to Actionable Intelligence
Automated protocol evaluation pipeline

STEP 1 Ingestion — Synthetic trial records stored in SQLite.

STEP 2 Evaluation — Visit telemetry checked against protocol rules.

STEP 3 Storage — Deviations linked to patient, site and visit.

STEP 4 Severity — Major, Minor, Administrative.

STEP 5 Risk Score — Weighted formula computes 0-100 site risk score.

STEP 6 AI Copilot — Database-grounded Q&A.

STEP 7 CAPA — Automated Root Cause & Preventive Action generation.

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Inside TrialGuard AI
Full-screen web dashboard user journey

1. Dashboard — Macro trial oversight and critical site alert.

2. Sites Overview — Risk cards, risk scores, major deviation counts and risk drivers.

3. Deviation Details — Protocol violations with patient/visit context.

4. AI Copilot — Example: "Why is SITE-003 high risk?" → 100.0/100 CRITICAL.

5. CAPA Generator — Root Cause Analysis & Preventive Action report export.

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IBM Bob as Our AI Development Partner
AI-assisted software development & grounded Copilot architecture

Architecture & Structure — React, FastAPI and SQLAlchemy decomposition.

Feature & API Implementation — Deviation detector and weighted risk scoring.

Debugging & Verification — Full-stack verification test scripts.

Documentation & Audit — Submission manifest validation and repository compliance audit.

Grounded flow: User Question → TrialGuard Copilot → SQLite DB → Grounded Context → watsonx.ai when configured / Deterministic Fallback → Response.

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Validated Working Prototype & Benchmark
Empirical verification results from local automated test suite

5 clinical trial sites monitored; 50 enrolled patients across 250+ scheduled visits.

SITE-003 dynamically computed at 100.0 / 100 — CRITICAL.

Protocol Deviation Detector: 100% detection accuracy across all 5 protocol rule types.

Database-Grounded AI Copilot: 100% telemetry verification pass rate.

CAPA Root Cause Analysis & PDF exporter verified.

Frontend build PASS; FastAPI backend tests PASS; repository validation workflow PASS.

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From Detection to Action
Transforming clinical trial risk oversight with actionable AI intelligence

Clinical Impact: Earlier risk identification; faster deviation review; explainable prioritization; structured CAPA support.

Future Roadmap: Medidata Rave & Veeva Vault connectors; clinical note NLP; HIPAA-compliant cloud & RBAC; validated GxP audit logging.

Disclaimer: TrialGuard AI is a prototype decision-support demonstration and is not validated clinical, medical, patient-safety, or regulatory software.

TrialGuard AI — "Detect Earlier. Prioritize Smarter. Act Faster."

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